

QT Prep: Dice Expectation, Variance, and Max

Part A: $E[X]$ and $\text{Var}(X)$ for a fair 6-sided die

Let X be the outcome of rolling one fair 6-sided die, so $X \in \{1, 2, 3, 4, 5, 6\}$ with $P(X = k) = \frac{1}{6}$.

Expectation

$$E[X] = \sum_{k=1}^6 k \cdot \frac{1}{6} = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = \frac{7}{2} = 3.5$$

Variance

Where I slipped: forgot the definition. The identity is

$$\text{Var}(X) = E[X^2] - (E[X])^2.$$

Not $E[(X - E[X])^2]$ computed term-by-term (that works but is slower). Also not $E[X]^2 - E[X^2]$ (sign flipped).

Compute $E[X^2]$:

$$E[X^2] = \frac{1+4+9+16+25+36}{6} = \frac{91}{6}.$$

Then:

$$\text{Var}(X) = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{91}{6} - \frac{49}{4} = \frac{182}{12} - \frac{147}{12} = \frac{35}{12} \approx 2.9167.$$

Standard deviation: $\sigma = \sqrt{35/12} \approx 1.708$.

Worth memorizing: For a single fair n -sided die,

$$E[X] = \frac{n+1}{2}, \quad \text{Var}(X) = \frac{n^2-1}{12}.$$

Check: $n = 6$ gives 3.5 and 35/12. ✓

Part B: Expected value of the max of 3 dice

Let $M = \max(X_1, X_2, X_3)$ where the X_i are i.i.d. rolls of a fair 6-sided die.

Build the PMF via the CDF trick

The key move: the CDF of a max is easy because the max is $\leq k$ iff *every* die is $\leq k$:

$$P(M \leq k) = P(X_1 \leq k) P(X_2 \leq k) P(X_3 \leq k) = \left(\frac{k}{6}\right)^3 = \frac{k^3}{216}.$$

Then the PMF is the first difference:

$$P(M = k) = P(M \leq k) - P(M \leq k-1) = \frac{k^3 - (k-1)^3}{216}.$$

Tabulating:

k	$k^3 - (k-1)^3$	$P(M = k)$
1	1	1/216
2	7	7/216
3	19	19/216
4	37	37/216
5	61	61/216
6	91	91/216

Sanity check: $1 + 7 + 19 + 37 + 61 + 91 = 216$. ✓

Expectation

$$\begin{aligned} E[M] &= \sum_{k=1}^6 k \cdot \frac{k^3 - (k-1)^3}{216} = \frac{1 \cdot 1 + 2 \cdot 7 + 3 \cdot 19 + 4 \cdot 37 + 5 \cdot 61 + 6 \cdot 91}{216} \\ &= \frac{1 + 14 + 57 + 148 + 305 + 546}{216} = \frac{1071}{216} \approx 4.9583. \end{aligned}$$

Fair value: $E[M] \approx 4.958$.

Side observation: the distribution is very right-heavy

Even though $E[M] \approx 4.96$, most of the probability mass lives at 5 and 6:

$$P(M \geq 5) = \frac{61 + 91}{216} = \frac{152}{216} \approx 70.4\%.$$

$$P(M \leq 4) = \frac{64}{216} \approx 29.6\%.$$

Long left tail, mass concentrated at the top. This is why the fair value lands noticeably below 5 even though ≥ 5 is the modal region.

Part C: Making a market on it

Step 1: Anchor on fair value. Quote symmetrically around 4.958. A reasonable tight market might look like 4.85 / 5.05 (10 cents wide) or 4.90 / 5.00 if feeling confident.

Step 2: Check for skew. The question is whether the *distribution of outcomes around the quote* is symmetric enough that I don't need to shade the quote up or down.

Roughly, looking at what happens on either side of fair:

- Outcomes below fair (1–4): contribute about 29.6% of mass but with large downside magnitude.
- Outcomes at 5: $\approx 28.2\%$ of mass, small positive deviation from fair.
- Outcome 6: $\approx 42.1\%$ of mass, large positive deviation.

The win/loss counts around a market centered at fair value work out close to $\approx 48/52$ — **practically symmetric in P&L terms once you weight by magnitude**. The mass is skewed toward high outcomes, but the large-magnitude low outcomes (rolling a 1 or 2) balance it out, which is exactly what $E[M]$ being just under 5 tells you.

Takeaway: no need to shade the quote. Market both sides around 4.96, symmetric spread. The expected value does all the work of pricing the skew already — don't double-count it by also widening asymmetrically.

General patterns to remember

1. **Variance:** $\text{Var}(X) = E[X^2] - (E[X])^2$. Always.
2. **Max of i.i.d. via CDF:** $P(\max \leq k) = \prod P(X_i \leq k)$. Differentiate (or difference, for discrete) to get the PMF.
3. **Min of i.i.d. via CDF:** mirror trick — $P(\min > k) = \prod P(X_i > k)$.
4. **Fair market pricing:** quote around $E[\cdot]$ first; only skew the quote if there's a reason the expected P&L is asymmetric at the quote prices (inventory, adverse selection, fat tails). Mean already prices first-order skew.
5. **For general n dice, max formula:** $E[\max] = \sum_{k=1}^6 k \cdot \frac{k^n - (k-1)^n}{6^n}$. For $n = 3$: 4.958. For $n = 6$: ≈ 5.596 .